CLIENT	**:**	**OBEROI REALTIY**
SITE	**:**	**?, MUMBAI**
LOCATION	**:**	**MUMBAI, MHARASHTRA**
NO.	**:**	**-**
DOCUMENT NO.	**:**	**DBR-R00**

R00	28-06-2025	Design report	R.H.B	H.A.B	H.A.B
Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

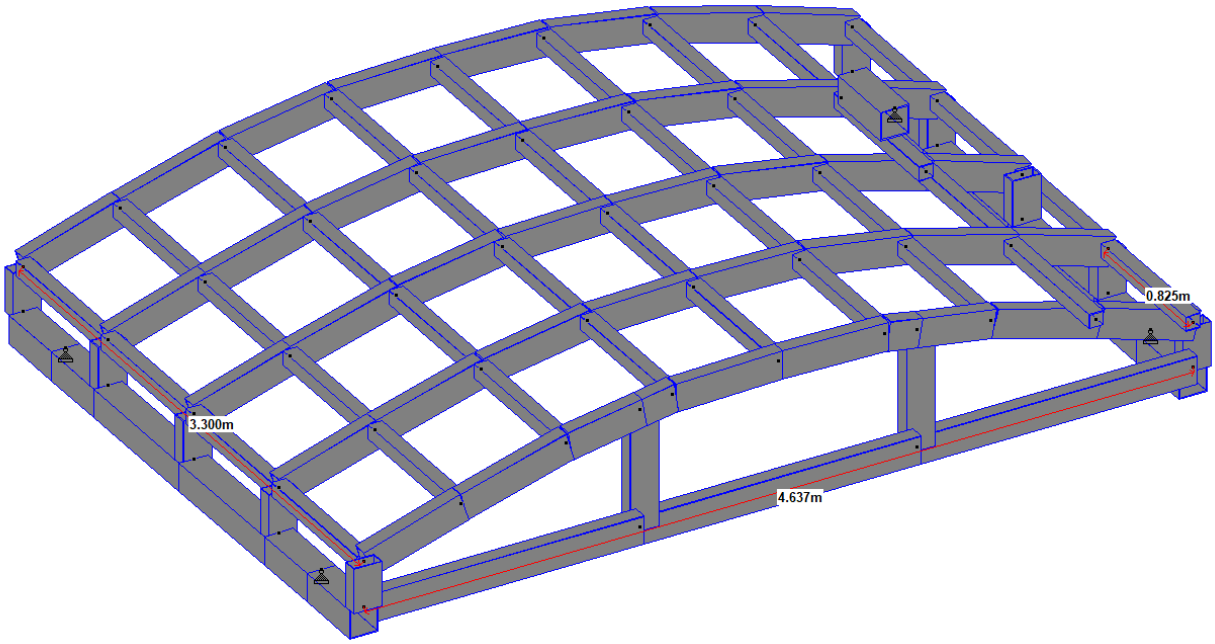
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1. Design Philosophy

This document pertains to structural design of **retractable roof**, Mumbai, Maharashtra at 215m from ground level. The overall size of the frame is 4.638x 3.4 x 0.7m high. The main sliding frame is of mild steel material. The wind load is worked out from IS 875 Part III. The various DL and LL acting at different points on frame are transferred to the nearest columns through the grid of beams.

The complete frame is modelled as 3D frame and analysed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output, the support reactions for various primary load cases are tabulated and the base plate, shear keys and foundation bolts are designed accordingly. Foundation design shall be done using appropriate load combinations for reactions given in this document.



2. DESIGN CODES AND STANDARDS

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
IS 875-Part III	For wind load calculation	--
IS 875-Part I	For dead load calculation	--

3. MATERIAL SPECIFICATION FOR CONSTRUCTION

Sr.no	Material	Specification	Minimum Strength
1	Mild steel	IS2062	Fy = 210 N/mm^2

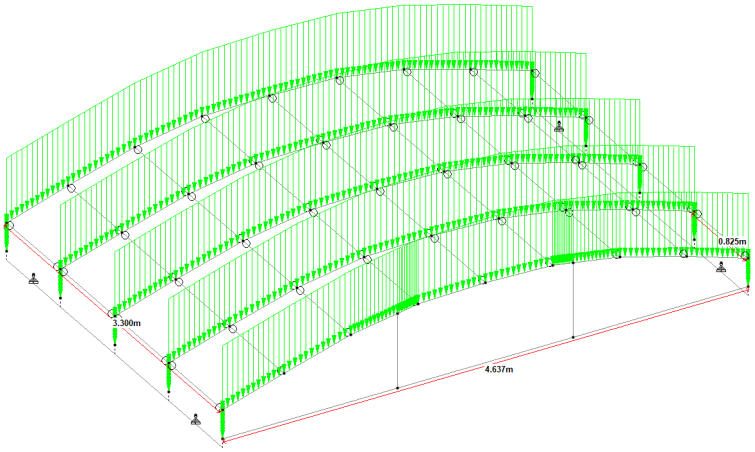
4. LOAD CALCULATIONS

4.1 DEAD LOAD

SELFWEIGHT FACTOR = 1

Dead load for glass/sheet assumed = 5 Kg/m^2 i.e. 0.05 kN/m^2

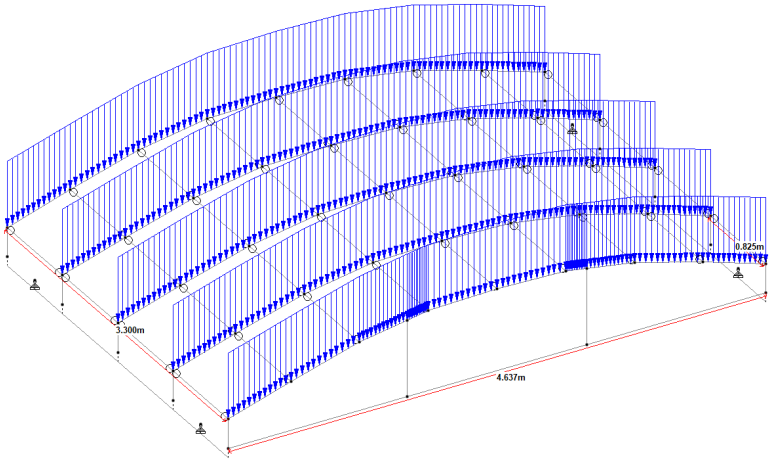
Member c/c spacing of 0.83m = 0.83x0.05 = 0.0415kN/m



4.2 LIVE LOAD

Live load assumed = 0.75 kN/m^2

Member c/c spacing of 0.83m = 0.83x0.75 = 0.6225kN/m

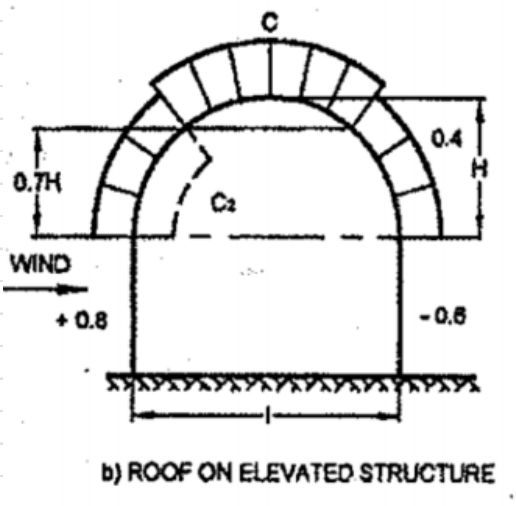


4.3 WIND LOAD

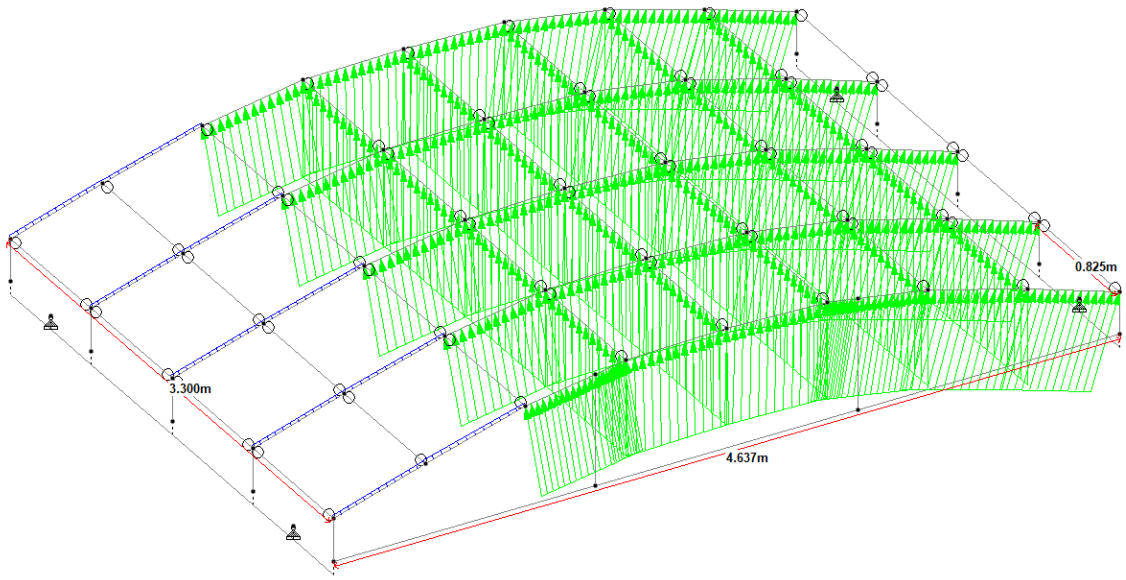
For wind load calculation, following IS code is referred: - "IS 875 Part III"

DESIGN WIND PRESSURE CALCULATION				
Sr no:	**Description**	**Value**	**Unit**	**Remarks**
1	V_b Basic Wind Speed	44	m/s	As per Apendix A (IS-875 Part 3)
2	K1 (risk coefficient)	1		For Design life span of 50 yrs. As per IS 875 Part III (table-1)
3	K2 (Terrain roughness and height factor) select terrain cat.	Catagory1		
	K2 for height 10m	10	1.05	As per IS 875 Part III (table-2), Cat-2, As per Ht
	K2 for height 15m	15	1.09	
	K2 for height 20m	20	1.12	
	K2 for height 30m	30	1.15	
	K2 for height 50m	50	1.2	
	K2 for height 100m	100	1.26	
	K2 for height 150m	150	1.32	
	K2 for height 200m	200	1.33	
	K2 for height 250m	250	1.34	
4	K3 (topography factor)	1		Clause 6.3.3 IS 875 part III
5	K4 (Importance factor for Cyclonic region.)	1.15		Clause 6.3.4 IS 875 part III
6	K_a (Area Averaging factor)	0.91		As per IS 875 Part III (table-4)
7	K_d (Wind Directionality Factor)	0.9		Clause 7.2.1 IS 875 part III
8	K_c (Combination Factor)	0.9		Clause 7.3.3.13 IS 875 part III
	Vz=	Vb*K1*K2*K3*K4		
	Vz (For 10m ht)	53.13	**m/s**	Vz= Vb*K1*K2*K3*K4
	Vz (For 15m ht)	55.154	**m/s**	
	Vz (For 20m ht)	56.672	**m/s**	
	Vz (For 30m ht)	58.19	**m/s**	
	Vz (For 50m ht)	60.72	**m/s**	
	Vz (For 100m ht)	63.756	**m/s**	
	Vz (For 150m ht)	66.792	**m/s**	
	Vz (For 200m ht)	67.298	**m/s**	
	Vz (For 250m ht)	67.804	**m/s**	
	Pz (For 10m ht)	1.69	kN/m2	$Pz = 0.6(Vz)^2$
	Pz (For 15m ht)	1.83	kN/m2	
	Pz (For 20m ht)	1.93	kN/m2	
	Pz (For 30m ht)	2.03	kN/m2	
	Pz (For 50m ht)	2.21	kN/m2	
	Pz (For 100m ht)	2.44	kN/m2	
	Pz (For 150m ht)	2.68	kN/m2	
	Pz (For 200m ht)	2.72	kN/m2	
	Pz (For 250m ht)	2.76	kN/m2	
	Ka*Kd*Kc =	0.74		**Use this value**
	Hence Use pd (For 10m Ht)	1.25	kN/m2	
	Hence Use pd (For 15m Ht)	1.35	kN/m2	
	Hence Use pd (For 20m Ht)	1.42	kN/m2	
	Hence Use pd (For 30m Ht)	1.50	kN/m2	
	Hence Use pd (For 50m Ht)	1.63	kN/m2	
	Hence Use pd (For 100m Ht)	1.80	kN/m2	
	Hence Use pd (For 150m Ht)	1.98	kN/m2	
	Hence Use pd (For 200m Ht)	2.01	kN/m2	
	Hence Use pd (For 250m Ht)	2.04	kN/m2	

Assuming roof of elevated structure,



L =	4.6
H =	0.7
H/l =	0.15
C=	-0.852
C2 =	0.055
Assuming center to center spacing of 0.83m	
Wind load in area	
C (kN/m)=	-1.44
C2(kN/m)=	0.09
Cpe =	-0.7
Assuming center to center spacing of 0.83m	
Wind load in area	
Cpe(kN/m)=	-1.18



5. Deflection check at building top in Z direction

Refer node no. –30 of staad file for load combination 202

Max. Deflection in Y dir. = 1.94 mm < L/250 = (IS 800-2007 table 6)

.... Hence ok.

6. PRIMARY LOADS & LOAD COMBINATIONS

6.1 PRIMARY LOADS

Load 1 Dead Load (DL)

Load 2 Live Load (LL)

Load 3 Wind Load Pressure (WLX)

Load 4 Wind Load Suction (WLZ)

Load 5 Wind Load Suction (WL-X)

Load 6 Wind Load Suction (WL-Z)

6.2 LOAD COMBINATIONS

Load Combination for strength:-

1.5 DL + 1.5 LL
1.5 DL + 1.5 WLX
1.5 DL + 1.5 WLZ
1.5 DL + 1.5 WL-X
1.5 DL + 1.5 WL-Z

Load Combination for serviceability:-

DL + LL
DL + WLX
DL + WLZ
DL + WL-X
DL + WL-Z
DL + 0.8 WLX
DL + 0.8 WLZ
DL + 0.8 WL-X
DL + 0.8 WL-Z